INVENTORY & SALES ANALYTICS

Supplier concentration, brand performance, pricing and inventory efficiency

ANALYTICS REPORT

A concise, decision-oriented review of the supplied notebook outputs and its ten visualisations.

2,645	2,301	87	67.46%
Rows in analysis table	Distinct product descriptions	Vendors represented	Purchases from top 10 vendors

Prepared from the uploaded Jupyter notebook and the ten supplied chart images. Findings are descriptive and should be validated against source tables before operational decisions.

Executive summary

The analysis combines product-level sales and purchase metrics with supplier and brand summaries. The clearest signals are supplier concentration, a set of high-margin/low-sales brands worth testing with targeted promotions, a lower observed unit purchase price for larger order groups, and inventory-value exposure that merits follow-up.

01 Supplier concentration	The top ten vendors account for 67.46% of total purchase dollars. Martignetti Companies alone contributes about 14.01%, followed by Diageo North America Inc. at 10.99%.
02 Brand opportunities	The notebook flags 66 descriptions in the bottom 15% of sales and top 15% of average profit margin. These are candidates for controlled promotion, placement or pricing tests—not automatic discounting.
03 Bulk purchasing	Mean unit purchase price falls from 16.87 for small orders to 8.33 for large orders in the notebook's order-size groups. The direction is clear, though product mix and order composition may also explain part of the difference.
04 Inventory exposure	The notebook calculates $133,889.37 as the total unsold-inventory value proxy and identifies vendors for review. This estimate depends on the notebook's quantity and purchase-price formula and should be reconciled to inventory records.
05 Margin comparison	The displayed chart output reports mean margins of 33.93% for the high-sales group and 34.55% for the low-sales group. Their plotted 95% confidence intervals overlap, so the chart does not show a clear separation in average margin.

Decision focus: protect continuity with major suppliers, investigate low-sales/high-margin products, test purchasing tiers where comparable products can be negotiated, and prioritize slow-moving stock before adding further inventory.

Scope & approach

Source and workflow

The notebook combines inventory, purchase, purchase-price, sales, end-inventory and vendor-invoice data into a final analysis table. The charting stage uses a filtered table where GrossProfit, ProfitMargin and TotalSalesQuantity are positive. The notebook reports 2,645 rows in the descriptive-statistics table and 2,301 distinct product descriptions in the brand summary.

Analytical methods shown

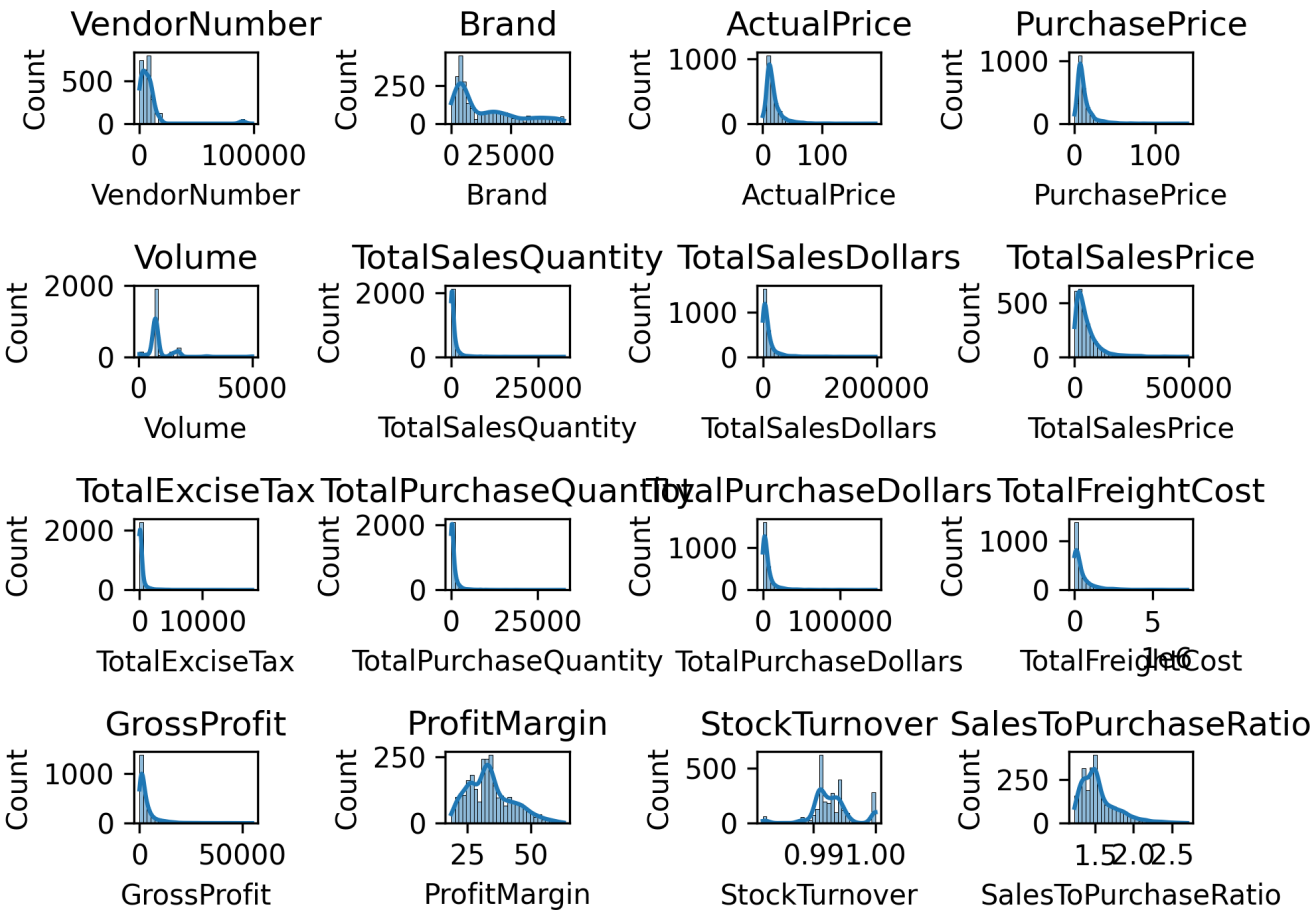
- Distribution histograms and boxplots for numerical fields.
- Top-category counts for vendor names and product descriptions.
- Pearson correlation matrix across numerical metrics.
- Brand targeting using the 15th percentile of sales and 85th percentile of average margin.
- Top-ten sales rankings and supplier purchase-contribution analysis.
- Order-size terciles, slow-stock screening, unsold-inventory valuation and 95% confidence intervals.

Important interpretation boundaries

- Correlations describe association, not causation; they do not prove that price changes cause revenue or margin changes.
- The final table appears to aggregate product/vendor metrics. Results depend on join keys, duplicate handling, time coverage and the definitions used for each metric.
- The field named TotalFreightCost has a very large scale in the distribution/boxplot. Verify its units and aggregation logic before using it in financial conclusions.
- The notebook's written confidence-interval paragraph does not match the values printed by its own executed cell. This report uses the executed chart/cell values and flags the mismatch later.

1. Numerical distributions

Figure 1 • Histograms and density curves for the analysis table's numerical fields.



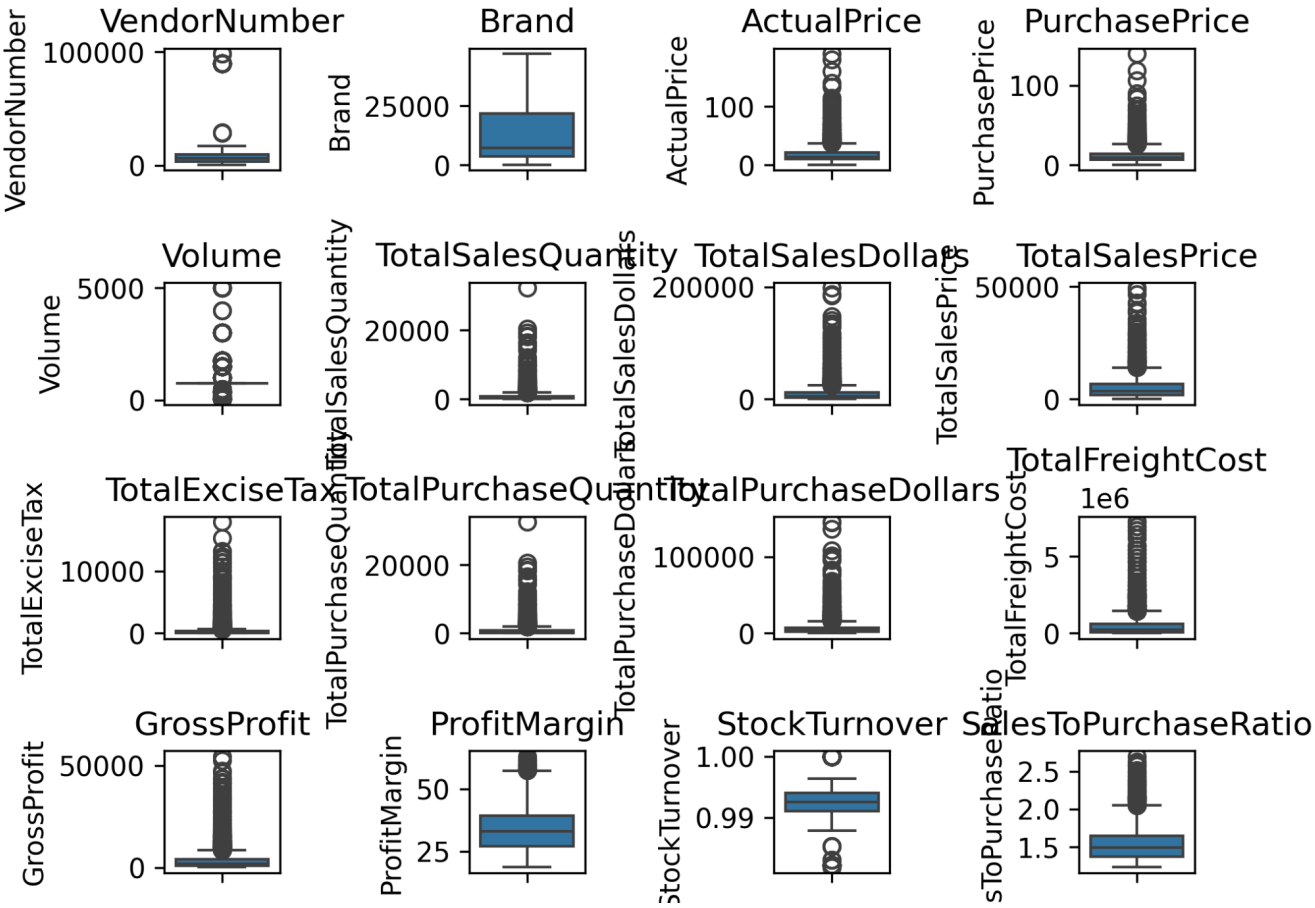
What the chart suggests

- Most monetary and quantity measures are strongly right-skewed: many products sit at lower values, while a smaller number have very large sales, purchase or tax totals.
- VendorNumber and Brand are identifiers, not continuous business measures; their histograms should not be interpreted as meaningful numeric distributions.
- ProfitMargin is more concentrated than the dollar/quantity totals, while StockTurnover is tightly clustered near 1.0.
- Consider log-scaled views for highly skewed monetary fields and inspect extreme observations before modelling or setting thresholds.

Interpretation note: These distributions describe the filtered analysis table, not necessarily individual transaction-level behaviour.

2. Outliers and spread

Figure 2 • Boxplots for the same numerical fields.



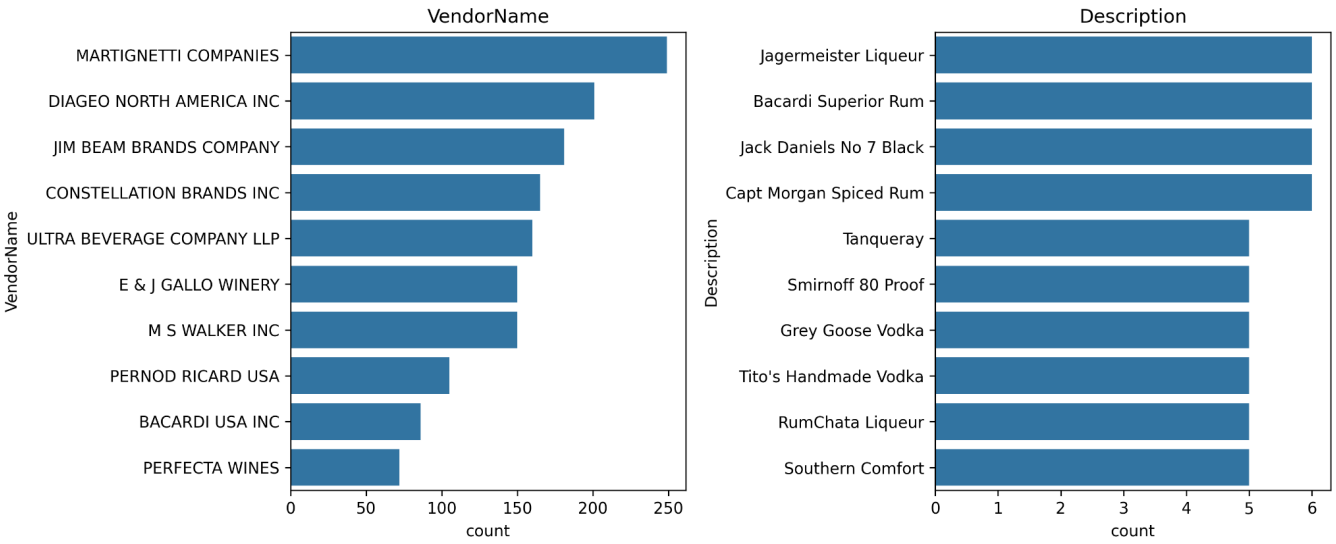
What the chart suggests

- Many monetary and quantity variables contain numerous high-end outliers, consistent with a long-tailed distribution and heterogeneous product scale.
- ActualPrice and PurchasePrice have high-price observations beyond the main interquartile range; these may be premium products rather than errors.
- StockTurnover has a narrow spread around 1.0, so small differences can look visually amplified.
- Outliers should be investigated by product, size and vendor before removal; valid high-volume or premium products may be commercially important.

Interpretation note: The TotalFreightCost scale is unusually large relative to other measures and warrants a units/aggregation check.

3. Vendor and product frequency

Figure 3 • Most frequent vendor names and product descriptions in the filtered table.

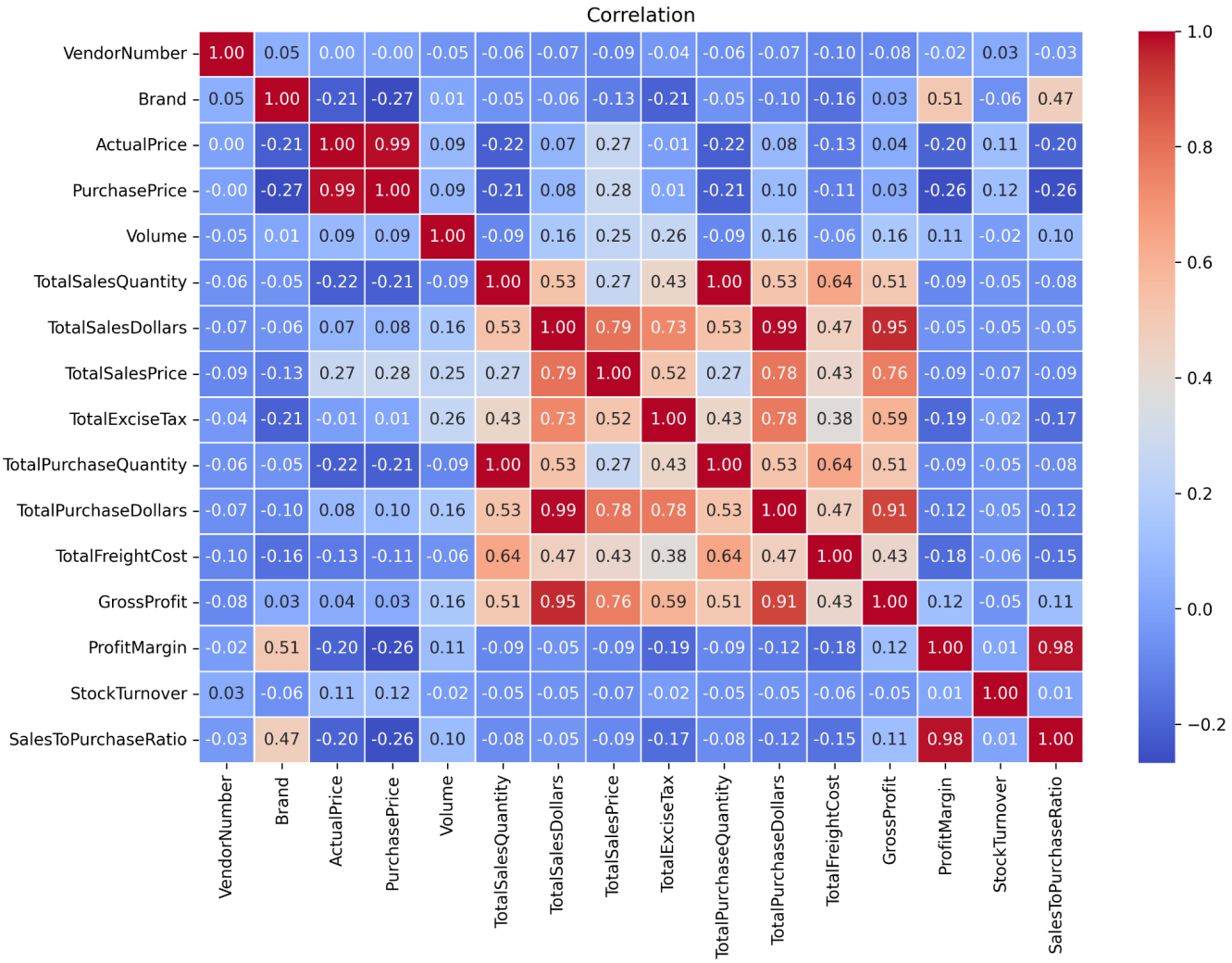


What the chart suggests

- Martignetti Companies appears most frequently among vendor records in this table, followed by Diageo North America Inc. and Jim Beam Brands Company.
- The leading descriptions have similar counts in the plotted top group; frequency is not the same as sales value or profitability.
- Use frequency as a coverage/record-volume indicator, then compare with dollar sales, purchase contribution and margin to understand business impact.

4. Correlation structure

Figure 4 • Correlations between numerical metrics.



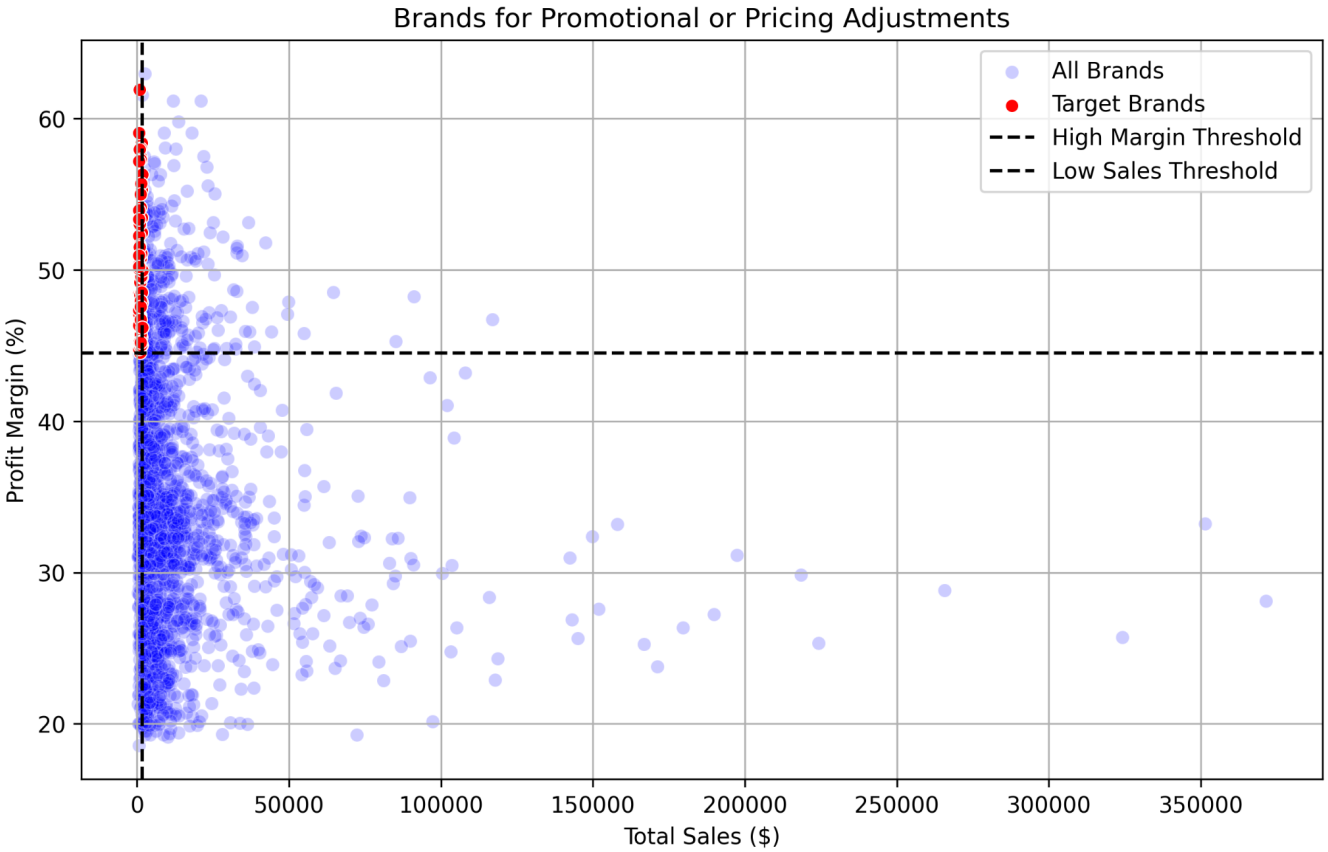
What the chart suggests

- ActualPrice and PurchasePrice are very strongly correlated (about 0.99), indicating that higher-priced items generally also have higher purchase prices.
- TotalSalesDollars is strongly correlated with GrossProfit (about 0.95) and TotalPurchaseDollars (about 0.99); larger-dollar products tend to drive larger dollar totals.
- TotalSalesQuantity and TotalPurchaseQuantity are almost perfectly correlated (about 1.00) in this aggregated table; confirm whether this reflects the construction of the final table before treating it as evidence of efficient turnover.
- ProfitMargin and SalesToPurchaseRatio are highly correlated (about 0.98), likely in part because the measures are mathematically related. ProfitMargin has weak negative correlations with ActualPrice and PurchasePrice.
- StockTurnover has near-zero correlation with GrossProfit and ProfitMargin, so turnover alone does not explain profitability in this sample.

Interpretation note: The notebook's written insight says PurchasePrice has weak correlation with sales dollars and GrossProfit; however, the displayed heatmap shows PurchasePrice's correlation with those variables is near 0.08 and 0.03 respectively, which is consistent with weak association.

5. Brands to consider for promotion or pricing review

Figure 5 • Brand sales against average profit margin; highlighted brands meet the notebook's targeting rule.



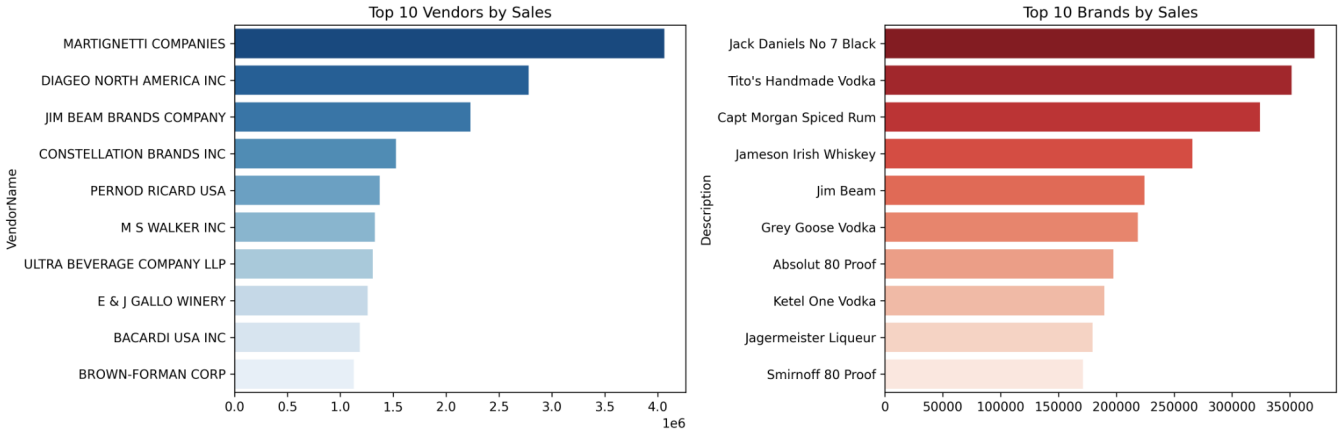
What the chart suggests

- The rule selects brands at or below the 15th percentile of sales (threshold: $1,648.90) and at or above the 85th percentile of average profit margin (threshold: 44.52%).
- The notebook identifies 66 product descriptions that meet both conditions.
- These products may have room for better visibility, assortment placement or targeted demand-generation because they combine relatively low sales with high average margin.
- Before applying discounts, test whether low sales reflect limited demand, distribution, seasonality, stock availability or niche positioning; discounting may erode the very margin that makes these products attractive.

Interpretation note: The scatter has a dense cluster near low sales values, so thresholds are a practical screening device rather than proof of a promotion effect.

6. Sales leaders: vendors and brands

Figure 6 • Top ten vendors and top ten product descriptions by total sales dollars.

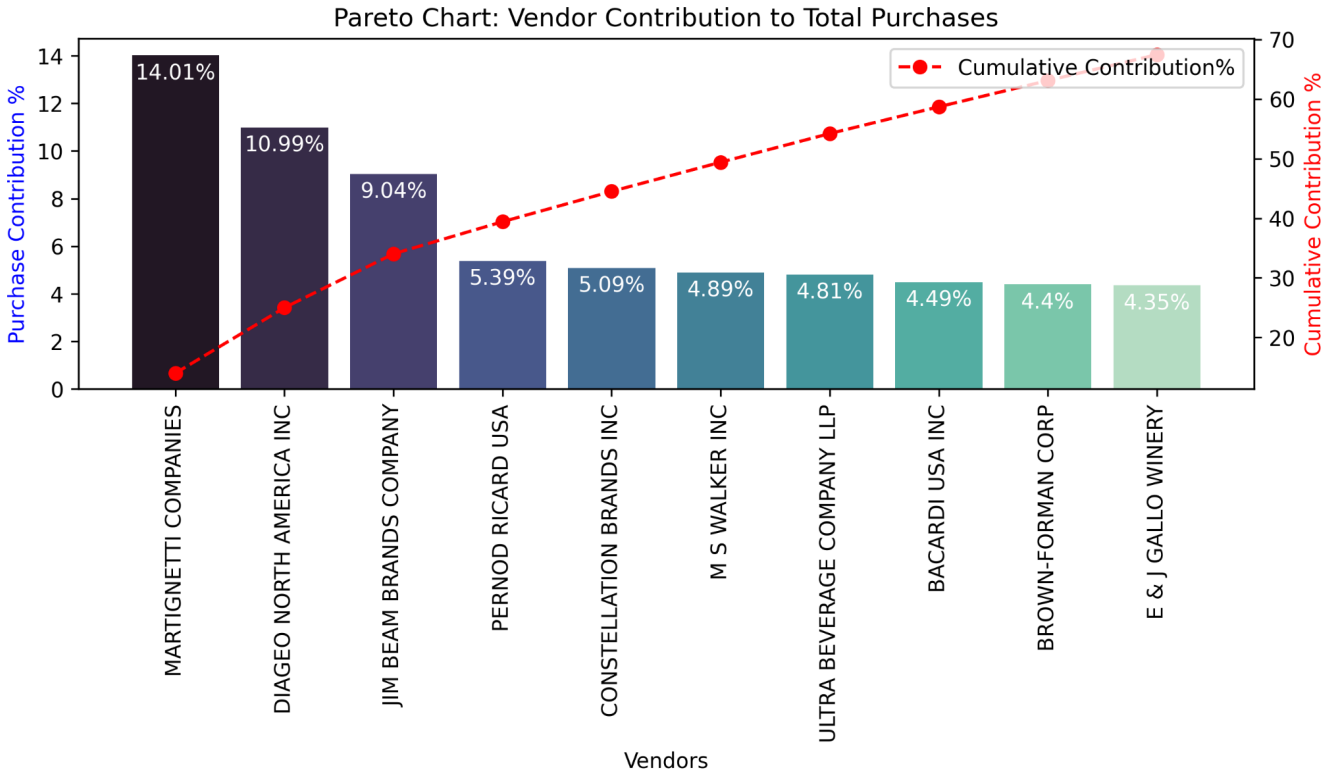


What the chart suggests

- Martignetti Companies is the largest vendor by sales dollars in this view, at approximately $4.06 million, followed by Diageo North America Inc. at $2.78 million and Jim Beam Brands Company at $2.23 million.
- Jack Daniels No 7 Black is the leading product description in the plotted brand ranking, followed by Tito's Handmade Vodka and Capt Morgan Spiced Rum.
- The vendor leaders overlap with the purchase-contribution leaders, suggesting that major suppliers are important on both supply and sales sides.
- Use the top-selling products to protect availability and monitor stockouts; use product-level margin and inventory ageing alongside sales to avoid over-prioritising revenue alone.

7. Supplier purchase contribution

Figure 7 • Pareto view of top-ten vendor purchase contribution and cumulative share.

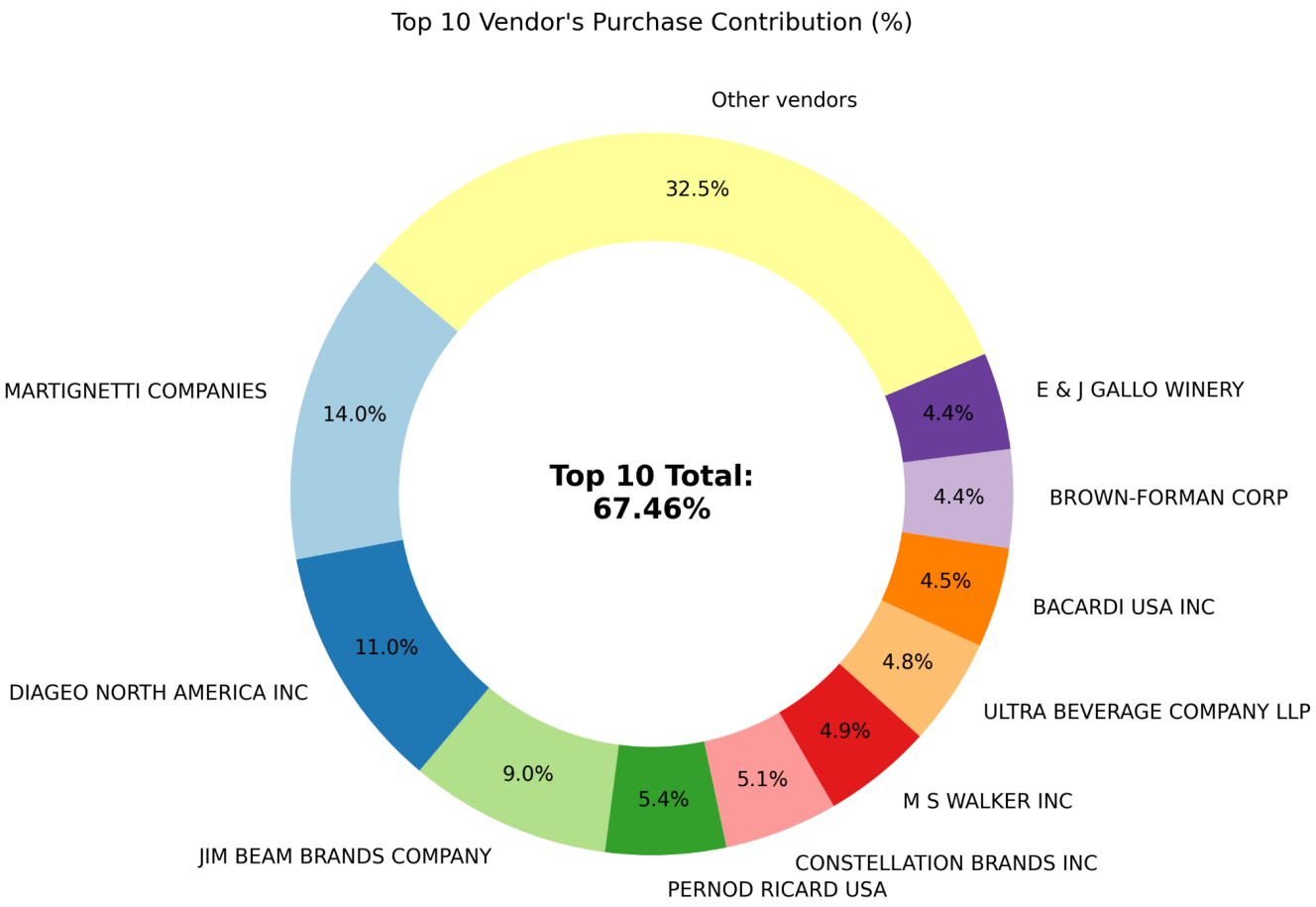


What the chart suggests

- Martignetti Companies contributes about 14.01% of total purchases, Diageo North America Inc. 10.99%, and Jim Beam Brands Company 9.04%.
- The top three suppliers account for about 34.04% of purchase dollars; the top ten account for approximately 67.46%.
- This concentration creates leverage for structured negotiations but also creates continuity risk if a major supplier has a disruption or changes terms.
- Maintain supplier-specific contingency plans, track lead times and service levels, and avoid interpreting high purchase share as poor performance by itself.

8. Purchase share at a glance

Figure 8 • Top-ten suppliers compared with all other vendors.



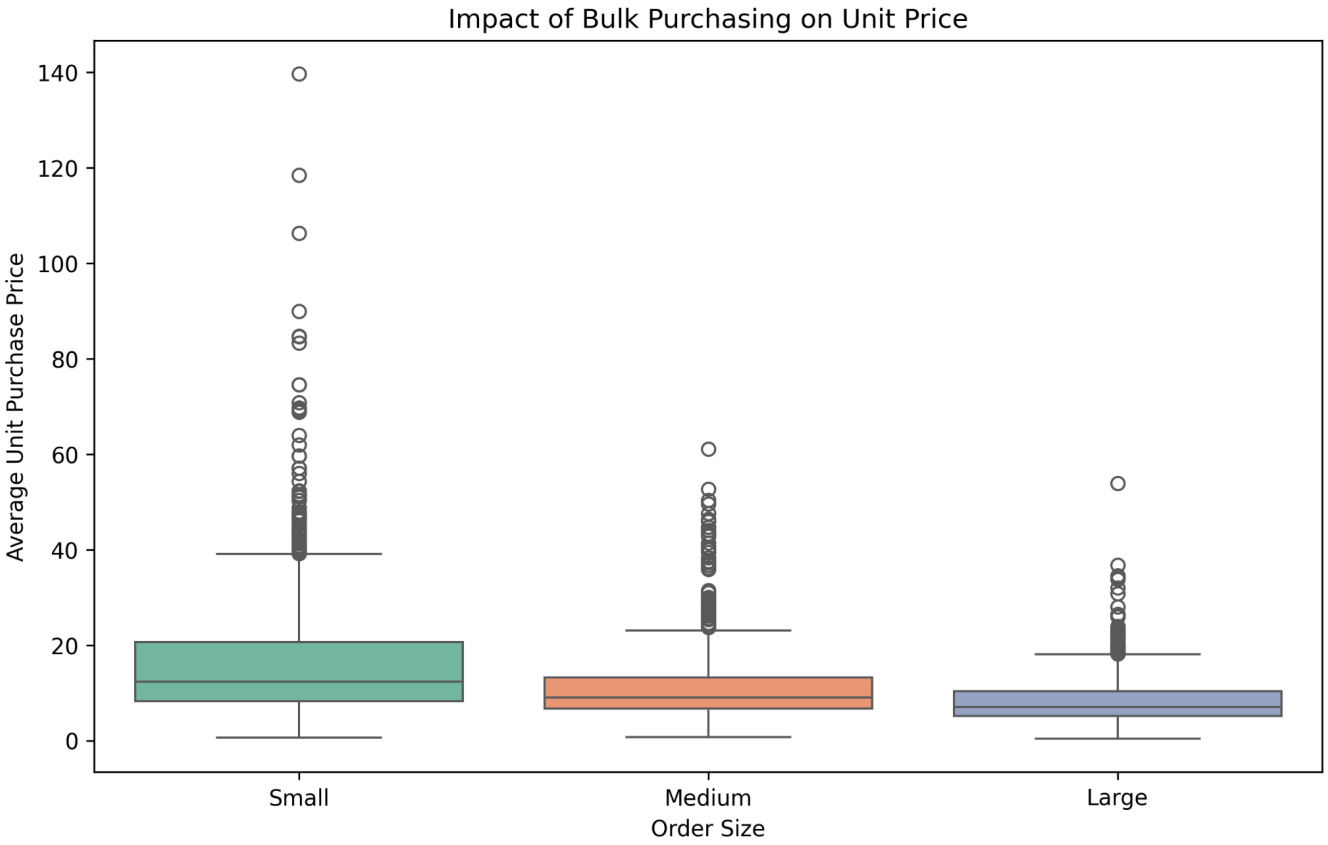
What the chart suggests

- The top ten suppliers together represent 67.46% of total purchases, leaving approximately 32.54% spread across other suppliers.
- Martignetti Companies is the single largest slice at roughly 14.0%; Diageo contributes roughly 11.0%.
- The chart reinforces the case for supplier relationship management and risk monitoring across the largest accounts, while preserving alternatives among smaller vendors.

Interpretation note: The Pareto and donut views communicate the same concentration result in complementary formats.

9. Bulk purchasing and unit price

Figure 9 • Unit purchase price across small, medium and large order-size groups.



What the chart suggests

- Mean unit purchase price is 16.87 for Small orders, 11.30 for Medium orders and 8.33 for Large orders.
- The mean for Large orders is about 50.6% lower than the Small-order mean using these printed group averages. The notebook's written note claims an approximately 72% reduction; that percentage is not supported by the means printed in the notebook output.
- The direction is consistent with volume-based pricing, but this is an observational comparison: different products, pack sizes or vendor mixes may contribute to the gap.
- For negotiation, compare like-for-like SKUs and pack sizes, quantify carrying/expiry costs, and set reorder quantities that capture price breaks without creating excess stock.

Interpretation note: The notebook's narrative says bulk pricing may support higher margins if inventory is managed efficiently; that is a plausible business hypothesis, not a causal result established by this chart.

10. Slow-moving stock & capital tied up

The notebook labels vendors with StockTurnover below 1 as potential slow-moving-stock candidates. Its ten lowest vendor averages range from approximately 0.987 to 0.990. The lowest listed values are Serralles Usa LLC (0.98697), ZORVINO VINEYARDS (0.98731) and Moonlight Meadery (0.98782).

The notebook then estimates unsold inventory value as (TotalPurchaseQuantity – TotalSalesQuantity) × PurchasePrice. Using that formula, the total is $133,889.37. The ten largest vendor-level values are led by Martignetti Companies ($19,169.12), Diageo North America Inc. ($15,459.03) and Jim Beam Brands Company ($12,068.92).

Recommended operational follow-up

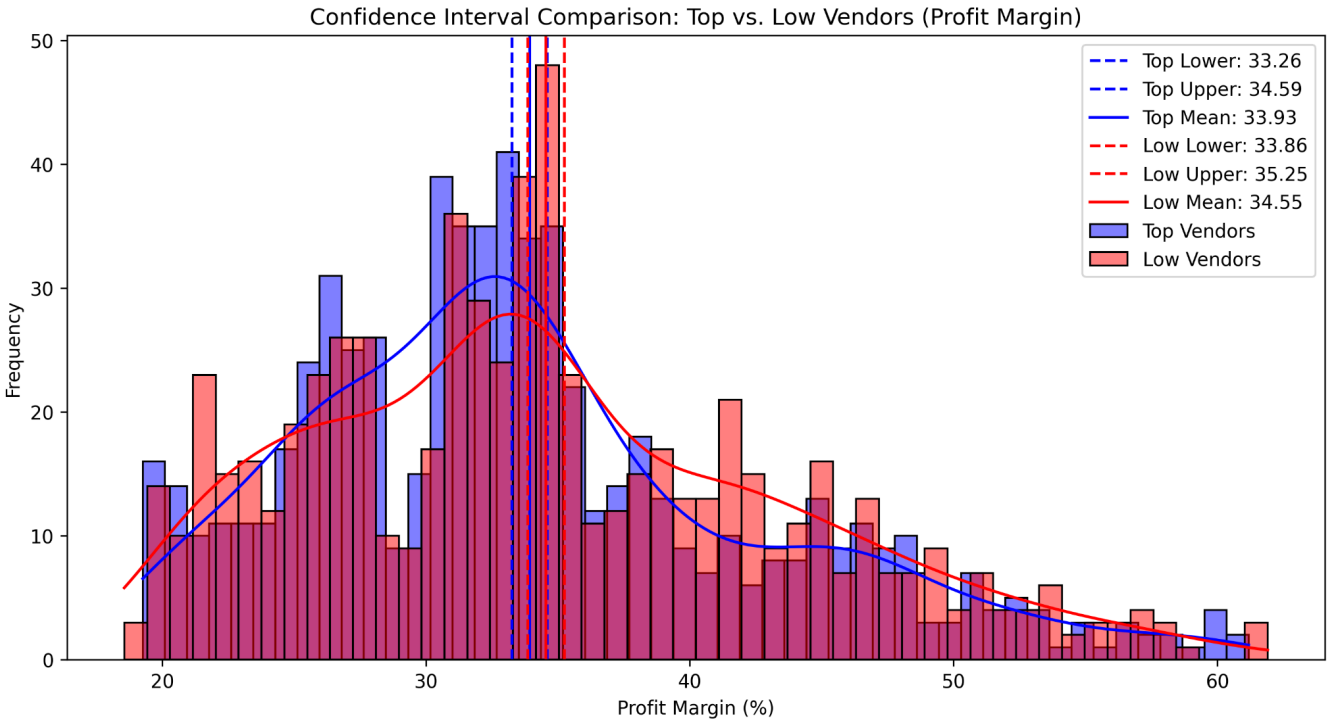
- Validate the stock-turnover definition and period; values clustered just below 1 may need business-specific thresholds.
- Reconcile estimated unsold quantities against opening stock, purchases, sales, transfers, returns and ending inventory.
- Prioritise ageing and sell-through checks for the vendors with the highest estimated unsold value.
- Consider markdowns, transfers, purchasing pauses or assortment changes only after confirming the stock is genuinely excess and saleable.

Caveat

The notebook's unsold-value calculation uses a simple quantity difference multiplied by PurchasePrice. If TotalPurchaseQuantity and TotalSalesQuantity are aggregated across different periods or are not directly comparable, this estimate may not equal accounting inventory value.

11. Profit margin: high-sales vs low-sales groups

Figure 10 • Histograms and 95% confidence intervals for products in the top and bottom sales quartiles.



What the chart suggests

- The executed notebook output reports a top-sales-group mean margin of 33.93%, with a 95% confidence interval of 33.26%–34.59%.
- The low-sales-group mean is 34.55%, with a 95% confidence interval of 33.86%–35.25%. The intervals overlap, so this plot alone does not establish a statistically clear difference in mean margin.
- The low-sales group has a slightly higher sample mean margin in this output, so lower sales performance may be driven by demand, distribution, price positioning, availability or product mix rather than margin alone.
- For high-performing vendors/products, protect volume while looking for cost optimisation. For low-performing ones, investigate sales drivers before reducing price.

Interpretation note: The notebook's Markdown inference states different values (top CI 32.90%–33.89%, low CI 33.13%–33.94%, means 33.39% and 33.53%). These conflict with the executed output and plotted labels; verify the notebook was re-run consistently before presenting the estimates externally.

Recommendations

Prioritise actions that can be tested and measured rather than treating descriptive patterns as definitive causes.

Supplier risk & negotiation	Build a quarterly supplier scorecard for the top ten by purchase share, covering fill rate, lead time, price changes, returns and continuity risk. Use concentration data to prioritise account reviews.
Brand growth experiments	Select a small pilot from the 66 high-margin/low-sales descriptions. Test visibility, placement or targeted promotions against a comparable control group and track incremental sales and gross profit.
Like-for-like bulk analysis	Compare unit purchase price for the same SKU, pack size and supplier across order quantities. Include freight, storage, cash tied up and potential obsolescence before changing reorder policies.
Inventory clean-up	Reconcile the $133.9k estimate to inventory records, then rank actual on-hand stock by ageing, sell-through and recoverable value. Pause replenishment for confirmed excess stock.
Metric validation	Audit the definitions of TotalFreightCost, StockTurnover, ProfitMargin and SalesToPurchaseRatio, including aggregation level and units. Document formulas in the report or data dictionary.
Statistical reporting	Re-run the confidence interval cell after the final filtering/aggregation steps and update the Markdown narrative so values, figure labels and interpretation agree.

Suggested next-step KPI set

Supplier concentration • on-time/in-full delivery • unit purchase price by SKU • sales and gross profit by product • inventory ageing • stockout rate • sell-through • promotion lift • gross profit after discount.

Appendix: notebook inferences and validation notes

The report preserves the notebook's main written interpretations while making numerical discrepancies explicit.

Correlation insights recorded in the notebook

- PurchasePrice has weak correlations with TotalSalesDollars and GrossProfit, suggesting that price variations do not significantly impact sales revenue or profit.
- The notebook interprets the strong relationship between total purchase quantity and total sales quantity as evidence of efficient inventory turnover. Because the correlation is almost exactly 1.00, the aggregation logic should be checked before making that operational conclusion.
- The notebook proposes that a negative relationship between ProfitMargin and TotalSalesPrice may reflect competitive pricing pressures. This is a possible explanation, not demonstrated causality.
- The notebook notes that StockTurnover has no meaningful correlation with GrossProfit or ProfitMargin.

Bulk-purchase inference recorded in the notebook

- The notebook states that large orders receive the lowest unit price and that this may support higher margins when inventory is managed efficiently.
- The notebook's written estimate of a ~72% reduction differs from the displayed group means (16.87 vs 8.33, a reduction of about 50.6%). This report uses the executed group means.

Confidence-interval inference recorded in the notebook

- The notebook's narrative says the top and low vendor confidence intervals overlap slightly and that their mean margins are similar.
- The executed cell/figure values are Top: mean 33.93%, CI 33.26%–34.59%; Low: mean 34.55%, CI 33.86%–35.25%. The Markdown narrative lists different values, so it should be updated.

Limitations

- No causal design or controlled experiment is shown; promotional, pricing and bulk-purchase suggestions should be tested.
- The provided images and notebook outputs are descriptive snapshots. The underlying CSV files were not separately audited as part of this report.
- Currency symbols are shown as $ where the notebook's chart context uses dollar labels; the dataset's currency/unit convention should be confirmed.

End of report • Prepared from the supplied notebook and chart images